Forums, Feedback, and Two Kinds of AI: A Selective History of Learning @ Scale

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ABSTRACT

Since the beginning of the Learning @ Scale conference in 2014, research has focused on a variety of topics. In this paper, we look at trends in four specific topics over time to identify how focus on these research directions has changed over the past 10 years: discussion forums, AI and machine learning, accessibility and inclusivity, and peer review. We find that interest in discussion forums has remained relatively steady, while interest in artificial intelligence and accessibility & inclusivity has risen. Interest in peer review, by contrast, has waned considerably. These findings are based on a an analysis of all 562 total Learning @ Scale papers from 2014 through 2023, including full papers, works in progress, and short papers.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; **Artificial intelligence**; • **General and reference** → **Surveys and overviews**.

KEYWORDS

peer review, artificial intelligence in education, machine learning in education, discussion forums, accessibility and inclusivity

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1 INTRODUCTION

Building off of the momentum surrounding MOOCs, the ACM Learning @ Scale conference launched in 2014. From the beginning, the conference has covered a wide variety of topics, but building on the foundation of MOOCs opened up specific possibilities in a number of areas. One of the missions of MOOCs has been to increase access to quality education, and so its role in increasing accessibility and inclusivity coalesced as an early driving factor. Given that MOOCs build on a model that eschews traditional student:teacher or student:teaching assistant ratios, mechanisms for supplying feedback and community have been a focal point, leading to an emphasis on peer review and discussion forums. And as artificial intelligence and machine learning approaches have risen in prominence, MOOCs and other at-scale learning environments have proven to be fruitful arenas in which to test out these initiatives and put them into practice.

A decade has passed since the conference began, so it is prudent to look at the extent to which these early research agendas have developed. In this research, we chart the prominence of papers on these four topics over time. These topics are not selected as *the* most important topics covered by the conference, but rather as topics that are of interest to the community. With the hype [14] surrounding MOOCs dying down, have these points of emphasis changed notably? How have they intersected with new environments like at-scale degree programs and informal learning environments? To answer these questions, we coded all 562 papers from the history of Learning @ Scale for whether these topics were central to the paper's goal, then charted the prevalence of each topic over time.

2 RELATED WORK

Past analyses have similarly attempted to capture the history of the Learning @ Scale conference. First, Kross & Guo in 2018 [40] surveyed the first four years of the conference, classifying 69 full papers published at the time into three categories: Systems for Learning @ Scale, Interactions with Sociotechnical Systems, and Understanding Online Students. Second, Kulkarni 2019 [41] evaluated the first five years of Learning @ Scale with a view towards two design perspectives: scaling efficiency and scaling empowerment. For example, two initiatives described by Kulkarni 2019 regarding mindset interventions note the difference between these approaches—one expanded access to mindset [20] interventions by creating a web site where anyone can come any participate in such a program [71], while the other created a toolkit that can be shared with teachers to more deliberately incorporate such interventions into their own classrooms [6]. The first targets users directly, while the second targets the community as a whole. Third, Johanes [33] performed an epistemological analysis of the first five years of the conference. This analysis applied a handful of techniques for clustering and connecting the early years' contributions to one another and to other related fields such as human-computer interaction and social

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psychology. Finally, in 2023, a work-in-progress paper evaluated the frequency per year of papers during the first eight years belonging to a wide variety of categories, including the approach taken, the environment targeted, and the theme of the paper [19]. This analysis builds most closely on that 2023 analysis.

3 METHODOLOGY

There have been 562 total papers published at Learning @ Scale from 2014 through 2023; 323 (57.5%) of these are full papers, while the remaining 239 are WIPs. Six coders participated in coding these papers. Four of the coders—all graduate students in computer science—each coded all 562 papers for whether they focused on a particular topic: discussion forums, peer review, AI & machine learning, and accessibility & inclusivity. Coders were instructed to concentrate on the *primary focus* of the paper rather than *all* themes present in the paper. This allowed for efficiency in coding—it was rarely necessary to read an entire paper to understand its primary focus—and it served as a guide for determining if a particular code should be applied. The topic codes were:

- **Discussion Forums:** One of the primary focuses of the research is the discussion forum as a tool (e.g., how people interact in forums, what they talk about, how much they interact). This includes papers where discussion forum data are used to draw conclusions and analyze other aspects of a course *if* the research focuses on using primarily discussion forum data. This does *not* include research where discussion forum data are used in conjunction with other data sources to draw conclusions or analyze aspects of a course.
- **Artificial Intelligence (AI) & Machine Learning (ML) Applications in Education:** One of the primary focuses of the research is the use of artificial intelligence and/or machine learning in educational contexts (e.g., their use in educational tools, how students use them for assignments, their role in education in general). This does *not* include research where AI/ML are simply used to analyze data or as a method of answering a separate research question.
- **Accessibility & Inclusivity:** One of the primary focuses of the research is expanding access to education. Examples may include research on increasing enrollment from underrepresented groups, making content more accessible for people with disabilities, or bringing education to geographical areas without much educational infrastructure.
- **Peer Feedback & Peer Review:** One of the primary focuses of the research is the implementation, application, or effectiveness of peer feedback or peer review.

The remaining two coders—experts in the field who have published several papers at Learning @ Scale over the years—coded a semi-random subset (60, just over 10%) of the 562 papers for all four research topics. The subset was semi-random (rather than fully random) to ensure there would be a minimum number of papers coded as "Yes" by the original coders in all four topics to facilitate the interrater reliability calculation. Interrater reliability was calculated for each of the four research topics between the two experts and the coder responsible for that topic, across the subset of 60 papers, using Fleiss's Kappa [23]. Fleiss's Kappa was chosen because we were interested in agreement between all three people (the two experts and the single coder), given that only the single coder's labels were applied to the entire dataset. Fleiss's Kappa was calculated at 0.904 for discussion forums, 0.428 for AI & ML in education, 0.662 for peer review, and 0.689 for accessibility & inclusivity. The lower value for AI & ML was identified to be due to the difficulty in differentiating machine learning as a simple data analysis technique compared to machine learning for active intervention or as a primary research objective. Its results are included here as-is, though further research is needed to more cleanly define and delineate different types of artificial intelligence interventions.

4 RESEARCH TOPICS

For clarity of analysis, in this section we present the results of each investigation followed by a discussion.

4.1 Discussion Forums

With regard to discussion forums, 5.2% of all Learning @ Scale papers focus on forums; the percentages of each year's papers that focus on this topic are shown in Figure 1 and have fluctuated between 2.7% and 7.1% over time. Over half (52%) of the papers about discussion forums feature research conducted in the context of MOOCs, and over a third (35%) focus on computer science as their academic subject matter. There is nearly an even split between discussion forum full papers (48%) and WIPs (52%), and we observed no notable trends over time related to these two types of papers.

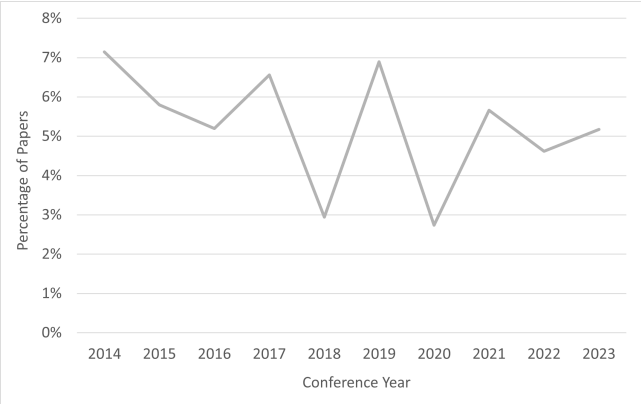

Figure 1: Percentage of Learning @ Scale papers about discussion forums over time.

4.1.1 Discussion. Research about discussion forums has been consistently present at Learning @ Scale over the past decade, albeit not as frequently as other topics. This seems counterintuitive, as not only do discussion forums offer various benefits for facilitating at-scale education (for example, allowing for peer learning and advising, and fostering community among students), but they also serve as a built-in repository of passively-collected student data that can—in theory—easily be used for research. In our review, we did indeed encounter many papers that analyzed discussion forum data. However, as explained previously, research on "discussion forums" was defined such that papers would be excluded from our

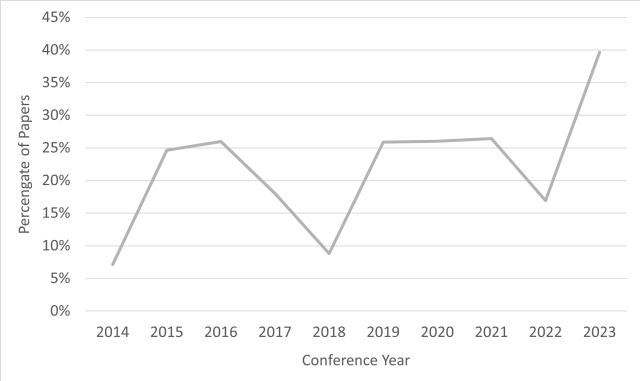

Figure 2: Percentage of Learning @ Scale papers about AI/ML in education over time.

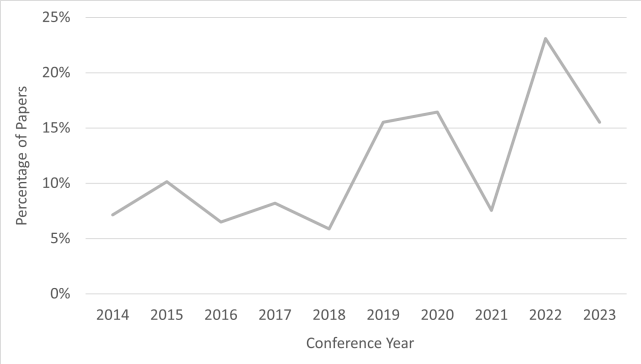

Figure 3: Percentage of Learning @ Scale papers about accessibility/inclusivity over time.

analysis if they approached discussion forums as a means of studying other aspects of at-scale education (a deliberate choice meant to help us focus our analysis).

Unsurprisingly, most research focusing on discussion forums was conducted in the context of MOOCs, where forums are perhaps most heavily utilized. Although we did not observe any trends over time in discussion forum research, we noted a variety of focus areas in these papers, including: student engagement [5, 50, 56, 72]; peer learning and advising [4, 18, 28]; automated post recommendation [8, 31, 32]; and student modeling and behavior [29, 58, 73].

4.2 AI and ML in Education

23.8% of all Learning @ Scale papers focus on applications of AI or ML in education; the percentages of each year's papers that focus on this topic are shown in Figure 2. These papers have increased in frequency over time from 7.1% of papers published in 2014 to 39.7% of papers published in 2023 (the highest across all years).

While we did not observe any clear trends in paper type year-to-year, we did observe that across the first five Learning @ Scale conferences (2014-2018), only 34.6% of AI/ML papers were full papers. In the most recent five conferences (2019-2023), however, 81.7% of AI/ML papers were full papers. We noted previously as well that 30% of papers coded with "Other" Research Contexts were AI/ML papers; 65% of AI/ML papers in turn are coded as arising from "Other" Research Contexts.

4.2.1 Discussion. Nearly a quarter of all papers at Learning @ Scale have focused on AI and ML and their applications in education, and we observed an increase in such papers over time, particularly at the 2023 conference, highlighting that this trend does not show signs of waning. In the earlier years of the conference, papers on this topic examined its role in grading [17, 25, 42–44, 47, 55, 65, 67, 74], intelligent tutoring systems [2, 16, 22], and content generation [10, 34, 53, 63, 66, 68, 69]; papers in later years examined the role of AI and ML in providing feedback [7, 13, 15, 24, 27, 37, 38, 45, 51, 57, 70] and maintaining academic integrity [3, 11, 26, 30]. Grades and course content are fundamental components of education, and papers on these topics in the beginning years of Learning @ Scale reflect an early focus on using AI and ML to assist with the mechanics of these components, transferring some of this burden away from instructors. Later research on feedback and academic integrity reflects expanding perspectives on the possibilities of using AI and ML in education to improve the student experience or boost the value and validity of a course or degree program.

The increase in the percentage of full papers (as opposed to WIPs) focused on AI and ML in education depicts a parallel narrative of the increased acceptance of these tools and recognition of their importance in education. This may also reflect the increased sophistication of these tools over the last 10 years, particularly as more data have become available for training AI agents. Based on the trends we observed, it is difficult to imagine areas of Learning @ Scale research that will not be influenced by AI and ML.

4.3 Accessibility and Inclusivity

12.3% of all Learning @ Scale papers focus on accessibility and inclusivity; the percentages of each year's papers that focus on this topic are shown in Figure 3. We observed an overall increase in the frequency of such papers from 7.1% in 2014 to 15.5% in 2023, with a maximum frequency of 23.1% in 2022. Overall, 63.8% of these have been full papers. Across the first five conferences, 50% of the papers on this topic were full papers, compared to 69.4% across the most recent five conferences.

4.3.1 Discussion. We encountered initial difficulty with coding papers for the topic of accessibility and inclusivity, leading to an unacceptably low Fleiss's Kappa. Even with an explicit definition for the code, there was more subjectivity in assessing papers for this topic than we had anticipated. Additionally, expanding access to education is almost inherently a result of scaling up education, and the argument could be made that most Learning @ Scale papers are, by definition, focused on accessibility and inclusivity.

However, after discussing the definition of this topic and agreeing on its parameters (which involved, among other things, excluding papers centered on general course design changes or automation), a subsequent coding of the papers yielded an acceptable value for Fleiss's Kappa. Although only 12.3% of all Learning @ Scale papers were categorized under our refined definition of "accessibility and inclusivity", the trend in frequency of such papers over time, as well as an increase in the number of full papers (as opposed to WIPs)

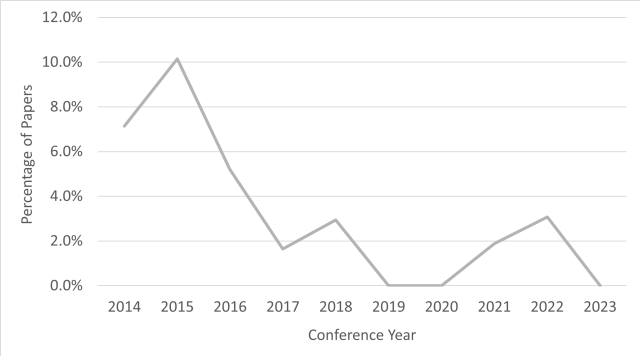

Figure 4: Percentage of Learning @ Scale papers about peer review/feedback over time.

in the conference's most recent five years, suggests an increasing interest in the topic. This may also reflect the field's growing maturity as efforts to expand access can be assessed empirically rather than remaining a long-term goal.

Papers on this topic have focused on accessibility of education for non-English speakers [59–62] and country specific research [1, 12, 52, 64] and, in more recent years, admissions [46, 54] and general design practices for inclusivity [9, 41, 49]. While a number of these papers focused on analysis of demographic or socioeconomic student data, we did not encounter a large number of papers testing interventions targeted at specific groups of students, such as [35, 48]. This is an area for growth for future Learning @ Scale research.

4.4 Peer Review and Peer Feedback

3% of all Learning @ Scale papers focus on peer feedback and peer review; the percentages of each year's papers that focus on this topic are shown in Figure 4. We observed a marked decrease in the frequency of such papers over time, from a peak of 10.1% of all papers in 2015 to 0% in 2023, with only three papers published on the topic in the five most recent conferences. With regard to paper type, there was nearly an even split between full papers (53%) and WIPs (47%) for all peer feedback/peer review papers, and we did not observe any trends over time. This trend likely coincides with the decrease in interest in MOOCs over time as peer review is often leveraged as a way of scaling open-ended assessment in MOOCs.

4.4.1 Discussion. Given the usefulness of peer review in scaling up education, surprisingly few Learning @ Scale papers have focused on these tools. In the context of the increase in research papers related to AI and ML—especially their role in feedback—this downward trend may reflect a move towards offloading the work of providing feedback to machines rather than students. Then again, the two are not mutually exclusive: we note a number of papers that examine how AI and ML can augment (rather than replace) peer feedback [13, 39, 42, 55, 67, 68, 75]. Many of these papers are from Learning @ Scale's earlier years, so there may be an opportunity for more current research on this topic. Other recent work has focused on including human experts in the evaluation process, similarly supplanting the need for peer grading [21, 36].

Deeper analysis of why minimal research has been conducted on the topic of peer feedback and peer review may be warranted. Are there obstacles to such research? Is there a lack of data? Are there implementation difficulties that make it not as beneficial to at-scale education as it would otherwise seem? Peer grading has often been seen as a solution to scaling challenges, but it is an opportunity for effective social learning on its own. Attempting to answer these questions may yield valuable insights into why this remains somewhat of a niche topic at Learning @ Scale.

5 LIMITATIONS

Qualitative coding is an imperfect science—even with well-defined codes, instructions, and training, it is nearly impossible to completely remove subjectivity. As mentioned, the coder responsible for coding papers related to accessibiilty and inclusivity took a broad view of this topic, which led to over half of the papers initially being assigned this topic code. With such a broad view, many—or even all—Learning @ Scale papers could be categorized as such, simply by virtue of being published at a conference centered around scaling up education and reaching new audiences. Further, the coder responsible for coding papers related to AI and ML in education, as well as the two experts, acknowledged difficulties with coding this topic, such as distinguishing research about a specific AI/ML educational intervention from research in which AI or ML was used to analyze data or answer a separate research question. With training, the accessibility and inclusivity coder was able to narrow their scope and re-code the papers, and the difficulties with coding the AI/ML topic did not have a significant effect on interrater reliability. Still, we acknowledge the potential for bias in qualitative coding.

6 CONCLUSION

In this work, we have selected four directions in Learning @ Scale research: discussion forums, peer review, AI & machine learning, and accessibility & inclusivity. We selected these directions not because they are *the* most important directions, but because they are individually interesting. Together, they formed over a quarter of research at the inaugural conference in 2014 based on our coding scheme, as well as over 60% in the most recent year—although that growth is largely driven by AI & machine learning in education growing from under 10% in 2014 to 40% in 2023.

The radical growth in AI research—which is seen in many other conferences and communities—risks turning Learning @ Scale into yet another AI in Education conference, which is already a saturated space with conferences like AI in Education and Intelligent Tutoring Systems, as well as related conferences like Educational Data Mining and Learning Analytics & Knowledge. In some ways, this parallels the early days of Learning @ Scale, when the conference was dominated by computer science education researchers and risked becoming "computer science education at scale". At the same time, there is an opportunity for a finer focus: many other communities focus on AI initiatives as technical endeavors, while Learning @ Scale is a community that puts these technologies into widespread practice. The conference has a route forward to distinguish itself and emphasize its role in bringing AI initiatives to the masses by focusing on research in authentic and at-scale contexts rather than those that merely have the *potential* to scale.

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